

EMPLOYMENT

Waymo Perception

Jun 2024—Present

Software Engineer, Semantics Foundation Team

- Built and deployed onboard vision-language models (VLMs) to enhance self-driving safety and robustness in long-tail scenarios.
- Led large-scale data mining, model evaluation, and system simulation to improve model reliability and real-world performance.
- Optimized model and system design for low-latency inference, achieving faster response times and improved overall efficiency.

EDUCATION

Cornell University

Aug 2019—Jun 2024

Ph.D., Computer Science

Advisor: Prof. Kilian Q. Weinberger

Shanghai Jiao Tong University

Sep 2015—Jun 2019

B.S., Computer Science

Advisor: Prof. Cewu Lu

PUBLICATIONS

- Learning 3D Perception from Others' Predictions
Jinsu Yoo, Zhenyang Feng, Tai-Yu Pan, Yihong Sun, Cheng Perng Phoo, **Xiangyu Chen**, Mark Campbell, Kilian Q Weinberger, Bharath Hariharan, Wei-Lun Chao
[ICLR 2025](#) [paper]
- CornerPoint3D: Look at the Nearest Corner Instead of the Center
Ruixiao Zhang, Runwei Guan, **Xiangyu Chen**, Adam Prugel-Bennett, Xiaohao Cai
[International Journal of Computer Vision, 2025](#) [paper]
- In-Vitro Blood Purification using Tiny Pinch Holographic Optical Tweezers Based on Deep Learning
Xiao Luo, Yu Ching Wong, **Xiangyu Chen**, Handong Tan, Weijia Wen
[Biosensors and Bioelectronics, 2025](#) [paper]
- Box Diffusion to Refine 3D Object Detections
Xiangyu Chen^{*}, Zhenzhen Liu^{*}, Siddhartha Datta, Katie Z Luo, Adhitya Polavaram, Yurong You, Yan Wang, Marco Pavone, Kilian Q Weinberger
[NeurIPS 2024](#) [paper]
- Reward Finetuning for Faster and More Accurate Unsupervised Object Discovery
Katie Z Luo^{*}, Zhenzhen Liu^{*}, **Xiangyu Chen**^{*}, Yurong You, Sagie Benaim, Cheng Perng Phoo, Mark Campbell, Wen Sun, Bharath Hariharan, Kilian Q Weinberger
[NeurIPS 2023](#) [paper]
- Waymax: An Accelerated, Data-Driven Simulator for Large-Scale Autonomous Driving Research
Cole Gulino, Justin Fu, Wenjie Luo, George Tucker, Eli Bronstein, Yiren Lu, Jean Harb, Xinlei Pan, Yan Wang, **Xiangyu Chen**, John D. Co-Reyes, Rishabh Agarwal, Rebecca Roelofs, Yao Lu, Nico Montali, Paul Mouglin, Zoey Yang, Brandyn White, Aleksandra Faust, Rowan McAllister, Dragomir Anguelov, Benjamin Sapp
[NeurIPS 2023](#) [paper]
- Learning Iterative Neural Optimizers for Image Steganography
Xiangyu Chen^{*}, Varsha Kishore^{*}, Kilian Weinberger
[ICLR 2023](#) [paper]

- Learning To Invert: Simple Adaptive Attacks for Gradient Inversion in Federated Learning
Ruihan Wu, **Xiangyu Chen**, Chuan Guo, Kilian Weinberger
[UAI 2023 \[paper\]](#)
- Ithaca365: Dataset and Driving Perception under Repeated and Challenging Weather Conditions
Carlos Andres Diaz, Youya Xia, Yurong You, Jose Nino, Junan Chen, Josephine Monica, **Xiangyu Chen**, Katie Z Luo, Yan Wang, Marc Emond, Wei-Lun Chao, Bharath Hariharan, Kilian Q. Weinberger, Mark Campbell
[CVPR 2022 \[paper\]](#)
- Hindsight is 20/20: Leveraging Past Traversals to Aid 3D Perception
Yurong You, Katie Luo, **Xiangyu Chen**, Junan Chen, Wei-Lun Chao, Wen Sun, Bharath Hariharan, Mark Campbell, Kilian Weinberger
[ICLR 2022 \[paper\]](#)
- Fixed Neural Network Steganography: Train the Images, not the Network
Varsha Kishore*, **Xiangyu Chen***, Yan Wang, Boyi Li, Kilian Weinberger
[ICLR 2022 \[paper\]](#)
- Low-Precision Reinforcement Learning: Running Soft Actor-Critic in Half Precision
Johan Bjorck, **Xiangyu Chen***, Christopher De Sa, Carla P. Gomes, Kilian Weinberger
[ICML 2021 \[paper\]](#)
- Train in Germany, Test in The USA: Making 3D Object Detectors Generalize
Yan Wang*, **Xiangyu Chen***, Yurong You, Li Erran Li, Bharath Hariharan, Mark Campbell, Kilian Weinberger, Wei-Lun Chao
[CVPR 2020 \[paper\]](#)
- Transferable Active Grasping and Real Embodied Dataset
Xiangyu Chen*, Zelin Ye*, Jiankai Sun, Yuda Fan, Fang Hu, Chenxi Wang, Cewu Lu
[ICRA 2020 \[paper\]](#)
- TendencyRL: Multi-Stage Discriminative Hints for Efficient Goal-Oriented Reverse Curriculum Learning
Chen Wang*, Junfeng Ding*, **Xiangyu Chen**, Zelin Ye, Jialu Wang, Ziruo Cai, Shixiang Gu, Cewu Lu
[IROS 2019 \[paper\]](#)
- Semantic Predictive Control for Interpretable and Efficient Policy Learning
Xinlei Pan*, **Xiangyu Chen***, Qizhi Cai, John Canny, Fisher Yu
[ICRA 2019 \[paper\]](#)

HONORS
AND
AWARDS

Outstanding Winner & Vilfredo Pareto Award

2018

Top 0.16%

Interdisciplinary Contest In Modeling

National Scholarship

2018, 2017, 2016

Top 0.2%

INTERNSHIPS

Waymo Research

May 2023—Aug 2023

Research Intern

Host: Yan Wang, Max Jiang, Dmitry Kalenichenko

Waymo Research

May 2022—Aug 2022

Research Intern

Host: Wenjie Luo

NEC Labs

Research Intern

May 2020—Aug 2020

Host: Pan Ji

TEACHING EXPERIENCE

CS358 Data Structure

Teaching Assistant

CS2110 OOP and Data Structures

Teaching Assistant

Sep 2016—Dec 2016

Shanghai Jiao Tong University

Aug 2019—May 2020

Cornell University

SERVICES

- Reviewer of CVPR 2019, 2022, 2023, 2024
- Reviewer of ECCV 2022
- Reviewer of ICLR 2025
- Reviewer of ICML 2025
- Reviewer of ICRA 2019, 2020, 2021, 2024
- Reviewer of IROS 2019
- Reviewer of NeurIPS 2023, 2024, 2025
- Reviewer of RSS 2019, 2020

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